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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester MCA (CBCS) Degree Examinations
 December 2024

22MCA344 – NETWORK AND CYBER SECURITY

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

	Unit – I	Marks	BT*	CO*	PO*
1. a)	Use the Caesar cipher to encrypt and decrypt the message " SECURITY" using the key (shift) value of 10	10	L*3	1	1
b)	Apply polyalphabetic cipher for the text "COMPUTER". Considering the key as "SCIENCES"	10	L3	1	1
2. a)	Apply Hill Cipher algorithm to encrypt the word "HI" and Decrypt it using the key and use a 2x2 matrix. $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$	10	L3	1	1
b)	Explain two common types of network attacks in cyber security.	10	L2	1	1
	Unit – II				
3. a)	In an RSA cryptosystem, B uses two prime numbers, 11 and 19, to generate the public and private keys. If the public key of B is 29, find the private key of B. Encrypt and decrypt the message M=5.	10	L3	2	1
b)	Explain Miller-Rabin algorithms.	10	L2	2	1
4. a)	Two parties, C and D, wish to establish a common secret key using the Diffie-Hellman key exchange technique. They agree on the modulus p=11 and the primitive root g=2. Party C chooses 4 as their secret and Party D chooses 7. What is the common shared secret key?	10	L3	2	1
b)	Explain Hash functions based on cipher block chaining with neat diagram.	10	L2	2	1
	Unit – III				
5. a)	Analyze the structure of an X.509 public key certificate and explain its general format.	10	L2	3	1
b)	Illustrate the message exchanges involved in Kerberos Version 4 and explain their purpose.	10	L2	3	1
6. a)	Compare and contrast the key differences between Kerberos Version 4 and Version 5.	10	L2	3	1
b)	Describe the process of X.509 public key certificate authentication and outline its key procedures.	10	L2	3	1
	Unit – IV				
7. a)	Classify and explain different types of cybercrimes.	10	L1	4	1
b)	Analyze the legal perspectives and frameworks related to cybercrimes.	10	L1	4	1
8. a)	Describe the concept of a cybercafe and explain the phenomenon of cyberstalking.	10	L2	4	1
b)	Explain the functioning of botnets and evaluate their impact on cybersecurity.	10	L2	4	1

P.T.O.

Unit – V

9.	a)	Explain DoS and DDoS attacks.	10	L2	5	1
	b)	Explain the following				
		i. SQL Injection and Buffer Overflow attack.				
		ii. Phishing and Password Cracking.	10	L2	5	1
10.	a)	Explain attacks on Wireless networks.	10	L1	5	1
	b)	Explain the following				
		i. Virus and Worms				
		ii. Trojan Horses and Backdoors	10	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
